

Panama



Income group: **High**



Number of questionnaires submitted: **13**



1 agency, 6 out of 6 managers,
6 out of 6 systems



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The AI and Data for Better Governance Survey

Artificial intelligence (AI) is rapidly enabling new possibilities to enhance governance—from designing and evaluating public policy to strengthening monitoring and transparency. In such a fast-moving field, it can be challenging for governments to evaluate their AI adoption compared to their peers, see what opportunities they might be leaving on the table, and strategize responses to common barriers. The World Bank Group aims to help governments meet this challenge by leveraging its position to offer a global picture of AI and data analytics use in government, as part of the *World Development Report 2026: Decoding AI*.

The AI and Data for Better Governance Survey was designed to accomplish the following:

- Measure the extent to which governments have already adopted AI
- Evaluate governments' institutional readiness and capacity to adopt AI in the future
- Document innovative approaches to applying AI and data analytics in public administration

The survey consisted of 13 questionnaires: one agency questionnaire focusing on AI adoption, strategy, infrastructure, and barriers from a whole-of-government perspective; six manager questionnaires focusing on ministries and agencies in six sectors (education, health care, public finance, procurement, taxation, and civil service); and six system questionnaires focusing on management information system (MIS) infrastructure, data readiness, and AI/machine learning capabilities in these same sectors.

The survey adopted a purposive expert-informant sampling strategy rather than a broad civil service census. Invitations were extended to all World Bank Group member states during the collection window from December 10, 2025, to May 15, 2026. Outreach and follow-up were conducted by the core survey team in Washington, DC, in collaboration with the World Bank Group's Country Management Units.

This economy brief presents findings from the survey along these three margins to help you compare your economy to the rest of the world. It was produced with the assistance of AI and coding tools to tailor survey results to your economy. Economy briefs are only being shared with economy counterparts involved in survey activities and World Bank staff. They summarize the responses to the survey provided by government counterparts, including your responses, and should be taken as descriptive summaries only, not as recommendations or proposed next steps. We do not intend to make the briefs publicly available, and they should be seen as working documents rather than formal publications.

Our aim is to make these survey data useful by giving our collaborators comparative information to inspire learning about other governments' efforts. We hope that, as stakeholders in this exercise, you feel empowered to use these data as the foundation for global collaboration to identify common challenges and share effective solutions. To that end, we are happy to put you in touch with others in the community so peer learning can continue.

What Does AI Adoption Look Like in Panama?

In Panama, the AI and Data for Better Governance Survey involved collaboration among several agencies, with the Autoridad Nacional para la Innovación Gubernamental responding as the central digital agency.¹ In total, 13 questionnaires were completed: the agency questionnaire, 6 manager questionnaires, and 6 system questionnaires (table 1).

Table 1. Manager and System Questionnaires, by Sector

	Civil Service	Education	Health	Procurement	Public Finance	Taxation
Manager	✓	✓	✓	✓	✓	✓
System	✓	✓	✓	✓	✓	✓

Source: AI and DfBG Survey. Check marks indicate completed survey instruments for this economy.

Drawing on data from the agency, manager, and system questionnaires, this section provides a snapshot of Panama's self-reported AI adoption, institutional readiness, and barriers.²

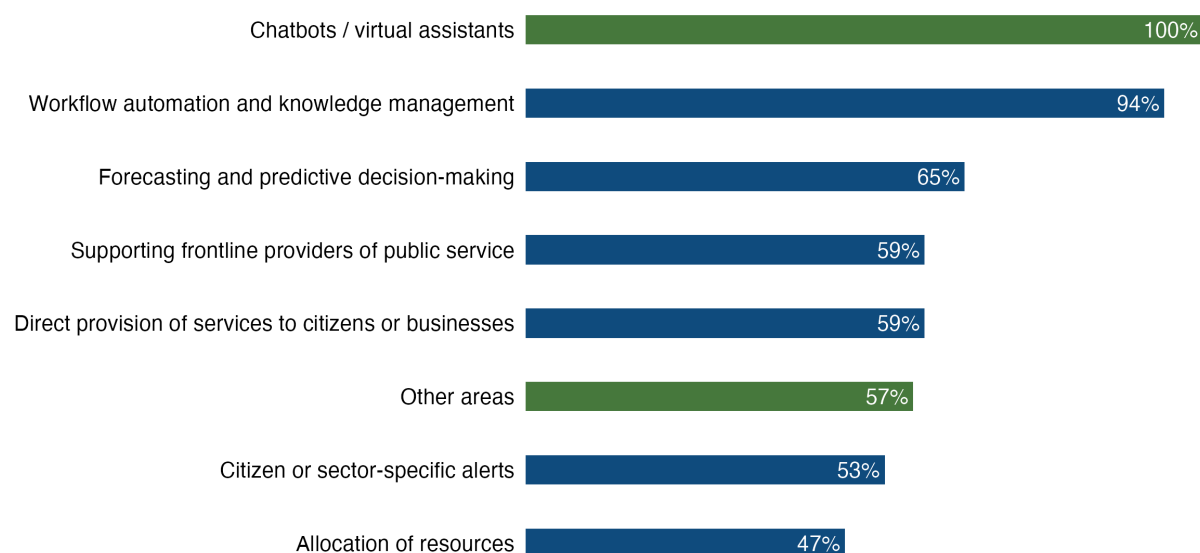
Current AI Adoption in Government

According to the survey, Panama's back-end AI use involves commercial AI tools embedded in internal workflows and used for routine tasks such as drafting, research, and coding support. Front-end AI use involves AI tools customized for public service delivery, such as tailored virtual assistants and workflow automation for service requests, embedded into routine citizen-facing processes. The government reported AI use in the following areas: chatbots and virtual assistants for public queries, and other areas not separately specified (figure 1).

¹Additional responding agencies included the Instituto Nacional de Estadística y Censo, the Ministerio de Educación, the Caja de Seguro Social, the Dirección General de Contrataciones Públicas, the Ministerio de Economía y Finanzas, and the Dirección General de Ingresos.

²AI-assisted tools were used to support the analysis and drafting of this report by summarizing survey responses and generating descriptive text. All content was reviewed, verified, and edited by the research team before publication to ensure its accuracy and consistency with the survey findings.

Figure 1. AI Use Cases (High-Income Economies)



Source: AI and DfBG Survey, agency questionnaire, question 9.

Note: Bars show the share of agencies in the same income group reporting each AI use case. Green bars indicate use cases this agency also reported, while blue bars indicate those it did not.

Possible AI uses that other governments have explored include direct service provision, budget forecasting, front-line provider support, and citizen alerts (figure 2). These example use cases are taken from around the world and may not be feasible or appropriate for every government; we include them as a source of inspiration only, not as concrete recommendations.

Figure 2. Examples of Other AI Use Cases (Global)

CITIZEN SERVICES AND E-GOVERNMENT		
Ukraine Diia AI AI assistant integrated in the national services portal to fulfill citizen requests and provide ready-made government services directly in chat.	United Arab Emirates TAMM Platform Consolidates services from 50+ entities into a single citizen and business journey with personalized recommendations and intelligent document triage.	Portugal Virtual assistant on GOV.PT Multilingual GenAI virtual assistant (12 languages) providing real-time guidance on 2,300+ public services.
Czechia AI-assisted pre-filling of public forms LLM and NLP-powered interface helping users complete electronic forms, reducing errors and improving accessibility.	Hungary AI-assisted service management Self-learning AI algorithm supporting back-office ticket handling in the centrally provided SaaS package for 3,200 municipalities.	Singapore AlBots Platform for public officers to create customized generative AI chatbots.
Australia AI for Operational Blueprint Helps frontline staff quickly find consistent guidance from long policy documents, reducing search time and errors.	Kazakhstan Eotinish AI AI within the e-appeals platform for managing citizen appeals.	Benin GPT-BJ Chatbot on the public services website that guides citizens to the right service based on their needs.

HEALTH

Chile

Analista Virtual – medical licenses
Virtual analyst at FONASA that reduces medical license processing times and shortens payment deadlines.

Chile

DART – ophthalmology AI
AI used by the Ministry of Health to prevent diabetic retinopathy and improve the availability of ophthalmologists.

Egypt, Arab Rep.

Breast Cancer MASA
AI diagnostic solution using computer vision to assist radiologists in breast cancer detection from mammograms.

New Zealand

Clinical consultation AI
Transcribes and summarizes clinical conversations, then automatically generates medical notes.

Türkiye

Turkish Brain Project
AI analyses MRI scans of brain tumor patients, identifying and classifying abnormal areas using advanced techniques.

Türkiye

Digital Eye – Mammography
AI decision support system analyzing mammography images to identify high-risk cases and priorities them for radiologist review.

Türkiye

Digital Eye – Liver Project
AI mass detection and classification integrated into the Ministry of Health's Cancer Early Diagnosis Screening Center.

EDUCATION

Argentina

Predicción de deserción escolar
Predictive model for school dropout prevention.

Uruguay

Ceibal virtual tutoring
AI-powered virtual tutoring for students in the Ceibal national education program.

Jordan

Siraj
AI-powered smart tutoring platform aligned with the Jordanian national curriculum, providing personalized adaptive learning.

PUBLIC SAFETY

Burkina Faso

Meta Oeil
Accident detection solution using urban surveillance video images.

Canada

Pre-load Air Cargo Targeting
AI system detecting risks in air cargo shipment data to prevent concealed explosive devices and threat items.

Tajikistan

Smart City cameras & traffic control
AI-powered smart city camera network for surveillance and traffic control.

Panama

Sistema de Alertas Tempranas – SINAPROC
Early warning and disaster simulation system for the National Civil Protection System.

South Sudan

INFLOW-AI – flood forecasting
Uses transformer models to forecast Nile flooding, enabling anticipatory funding and faster disaster response.

South Sudan

Disaster communication chatbots
UNESCO-trained AI chatbots disseminating disaster information, identifying safe locations, and improving disaster response efficiency.

JUSTICE

Australia

TM Checker
Free trademark availability checker using AI to identify similarity and distinctiveness issues, simplifying IP applications for small businesses.

Belgium

AI for social fraud detection
AI used by the National Social Security Office to detect social fraud.

Greece

Contracts' check in Cadastre
AI recognizes the content of contracts, identifies critical legal elements, and automatically generates recommendations to the supervisor.

Portugal

LIA
Chat-based search engine for legislation published in Portugal's Official Gazette, helping citizens navigate complex consolidated law.

TAX AND FINANCE

Austria

Grants4Companies

AI-powered tool to match companies with available grants.

Malaysia

Income tax fraud detection

AI system for detecting income tax fraud.

Mexico

Modelo de riesgos de Aduanas

AI risk model for customs inspections.

Uruguay

ANCAP fuel demand projection

ML-based projection of demand for gas oil, petrol, and LPG.

TRANSPORT AND INFRASTRUCTURE

Azerbaijan

Digital Twin for Transport

Digital twin model for transport infrastructure planning and management.

Costa Rica

Gestión del tránsito

AI-based traffic management system.

Jordan

AI for electrical load planning

AI predicting electricity loads and renewable energy generation, assisting grid operators at NEPCO.

Kenya

AI traffic systems

AI traffic management systems to reduce congestion in Nairobi.

Thailand

MOT Executive Dashboard

AI-powered executive dashboard for the Ministry of Transport covering passenger data.

AGRICULTURE AND ENVIRONMENT

Kenya

Chatbot and AI in agriculture

AI chatbot and agricultural tools including PlantVillage Nuru for crop diagnosis.

Kenya

CropGuard

Machine learning models for bird detection and crop protection for Kenyan farmers.

South Sudan

Smart Cassava

AI-powered mobile solution for crop diagnosis and severity analysis.

LANGUAGE AND TRANSLATION

Bhutan

English–Dzongkha machine translation

Machine translation for the public service delivery portal.

Burkina Faso

BeatMonitor

Used by the copyright office to detect exploitation of intellectual works on TV, radio, and in venues.

Cambodia

TranslateKH

Machine translation tool for English–Khmer and Khmer–English.

Cambodia

Sarika

Text-to-speech platform for the Khmer language.

Canada

GC Translate

Secure, real-time AI language tools trained on Canadian data, providing a self-service language hub for government employees.

Nigeria

N-ATLAS

Open-source multilingual LLM trained on Yoruba, Igbo, Hausa, and Nigerian-accented English.

Portugal

AMALIA

First LLM designed for European Portuguese, ensuring linguistic and cultural sovereignty.

Singapore

Transcribe

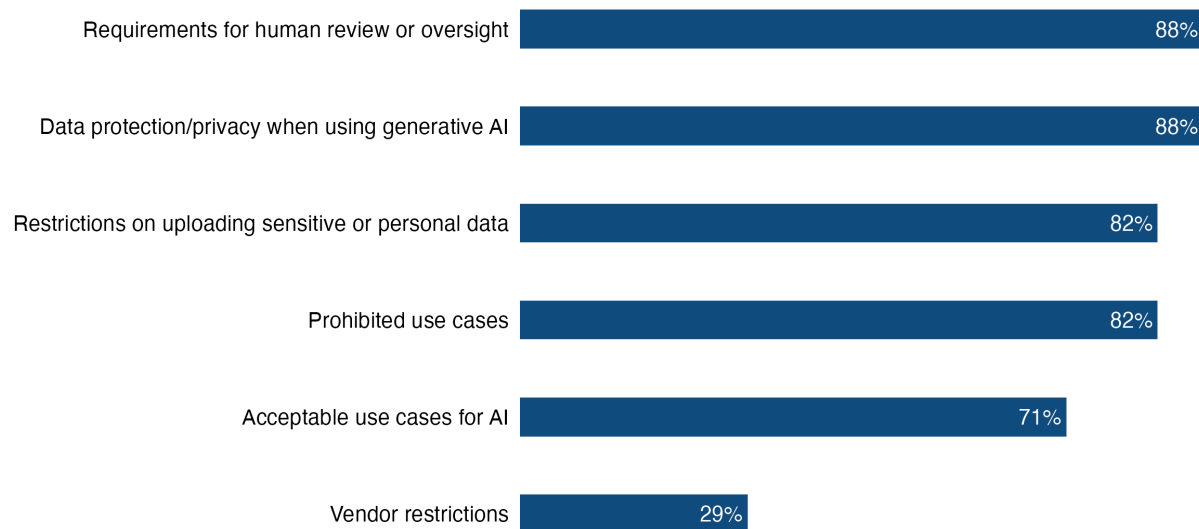
Speech-to-text platform for live transcription of meetings and source materials for public officers.

Source: AI and DfBG Survey, agency questionnaire, question 11.

In Panama, guidelines for the use of generative AI tools are currently under draft. Across governments at the same income level as Panama, the most commonly covered topics in AI guidelines are data protection and privacy when using generative AI and requirements for human review or oversight, each

addressed by 88 percent of economies in this income group. Restrictions on uploading sensitive or personal data and prohibited use cases follow, each covered by 82 percent, with acceptable use cases addressed by 71 percent. Vendor restrictions are the least commonly covered topic, addressed by 29 percent of economies in this income group (figure 3).

Figure 3. Topics Covered by AI Guidelines (High-Income Economies)



Source: AI and DfBG Survey, agency questionnaire, question 15.

Note: Bars show the share of agencies in the same income group whose AI guidelines cover each topic. Green bars indicate topics this government's guidelines also cover; blue indicate those they do not.

Institutional Readiness and Capacity for AI Adoption

Panama has a dedicated data science or analytics unit, though not one specifically focused on AI. Its national AI strategy is under development or in draft form, a level reported by 6 percent of peers, while 65 percent of peers have a strategy that is officially adopted and widely used across government. The government has no AI oversight body covering the public sector, whereas 94 percent of peers have one. It has a dedicated budget for data analytics, as do 71 percent of peers, but has not established a separate budget for AI. Some departments or units have purchased limited AI tool licenses.

Panama's AI and analytics training landscape varies considerably across sectors. Civil service and public finance report no analytics or AI literacy training program. Among the sectors that do have programs, education is the most developed: it offers advanced analytics training, with staff developing advanced analytical models, conducting complex data analysis and evaluation, and integrating data analytics into government workflows and decision-making processes. Its AI literacy training is fully institutionalized, mandatory or regularly offered across government, with moderate coverage accessible to several ministries or staff categories, and at an intermediate specialization level that includes preparing data for AI applications, using AI tools for analysis and visualization, and understanding basic model outputs and limitations. Procurement reports no analytics training program but does have an AI literacy training program, though its institutionalization, coverage, and specialization were not captured. Taxation has an analytics training program in place but its training level was not specified, and its AI literacy training status is uncertain (table 2).

Table 2. Analytics and AI Training by Sector (Panama)

	Civil Service	Education	Health	Procurement	Public Finance	Taxation
Analytics						
Analytics training program	No	Yes	Not sure	No	No	Yes
Analytics training level	N/A	Advanced	N/A	N/A	N/A	Not sure
AI						
AI literacy training program	No	Yes	Not sure	Yes	No	Not sure
AI training institutionalization	N/A	Fully	N/A	No response	N/A	N/A
AI training coverage	N/A	Moderate	N/A	No response	N/A	N/A
AI training specialization	N/A	Intermediate	N/A	No response	N/A	N/A

Source: AI and DfBG Survey, manager questionnaire, questions 25 to 32.

Note: “N/A” indicates the sector reported no program for that training category. “No response” indicates the sector reported having a program but did not answer the follow-up question. “Not sure” indicates the respondent was uncertain about the program’s existence.

Barriers to AI Adoption

Panama’s central digital agency rates data privacy, ethical, legal, security or social concerns as the sole major barrier, an assessment shared by 18 percent of peers who also rate it major. Rated moderate are data quality and availability, budget constraints, lack of talent to design and develop AI solutions, lack of staff skills to effectively use AI tools, and resistance to change and difficulty obtaining support from stakeholders. The government rates as minor barriers the lack of relevant policies, guidelines, frameworks or standards; limited political interest and leadership; legacy IT systems and interoperability issues; concern that AI tools may not work or meet needs; and lack of clear use cases for its organizations’ work. No barrier is rated not a barrier (table 3).

Table 3. Central Barriers to AI Adoption (Panama)

	Not a barrier	Minor	Moderate	Major
Lack of relevant policies, guidelines, frameworks, or standards				
Data privacy, ethical, legal, security, or social concerns				
Data quality and availability				
Budget constraints				
Lack of talent to design and develop AI solutions				
Lack of staff skills to effectively use AI tools				
Resistance to change and difficulty obtaining support from stakeholders				
Limited political interest and leadership				
Legacy IT systems / interoperability issues				
Concern that AI tools may not work or meet needs				
Lack of clear use cases for organization's work				

Source: AI and DfBG Survey, agency questionnaire, question 28.

Note: Blank rows indicate the agency responded “Not sure.” Colors reflect barrier severity from green (not a barrier) to red (major barrier).

Panama’s sector-level barrier lists show one item appearing in all six sectors: legacy IT systems and interoperability issues, making it the most pervasive constraint reported. Budget constraints appear in three sectors, namely health, procurement, and taxation. Insufficient technical expertise is cited in two sectors, civil service and public finance, while data quality and availability likewise appears in two, civil service and taxation. Data privacy, ethical, legal, security or social concerns are reported in education and taxation. Lack of relevant policies, guidelines, frameworks or standards appears in education and health. Several barriers surface in only a single sector: limited political interest and leadership is cited solely in procurement, and resistance to change and difficulty getting buy-in from staff or managers is cited solely in public finance (table 4).

Table 4. Sector-Level Barriers to AI Adoption (Panama)

	Civil Service	Education	Health	Procurement	Public Finance	Taxation
Lack of relevant policies, guidelines, and standards						
Data privacy, ethical, legal, and security concerns						
Data quality and availability						
Budget constraints						
Insufficient technical expertise						
Resistance to change and stakeholder support						
Limited political interest and leadership						
Legacy IT systems/interoperability issues						
Concern that AI tools may not work or meet needs						

Source: AI and DfBG Survey, manager questionnaire, question 20.

Note: Each sector identified up to three main barriers to AI adoption. Red cells indicate a barrier was selected.

System-level managers identified the following key barriers to use of sector MIS data for AI and advanced analytics (table 5).

Table 5. System-Level Barriers to AI Adoption (Panama)

	Civil Service	Education	Health	Procurement	Public Finance	Taxation
Data quality/completeness/timeliness						
Lack of interoperability/unique IDs						
Limited compute/cloud infrastructure						
Privacy/legal restrictions on data use						
Lack of standards for data handling and AI						
Skills gaps in MIS/IT team						
Budget constraints for upgrades/AI pilots						
Low buy-in from users/management						
Concerns about AI reliability/fit						

Source: AI and DfBG Survey, system questionnaire, question 18.

Note: Each system identified up to three main barriers limiting advanced analytics or AI. Red cells indicate a barrier was selected. Barrier categories differ from the manager questionnaire and reflect technical and infrastructure constraints.

In response to these barriers, Panama's central digital agency reported the following key actions for responsible AI implementation inside the government:³

1. AI framework for the national government

³Responses machine translated from original language.

2. National data governance; a committee is being formalized
3. Passage of a law to support innovation in the use of AI while protecting citizens' data

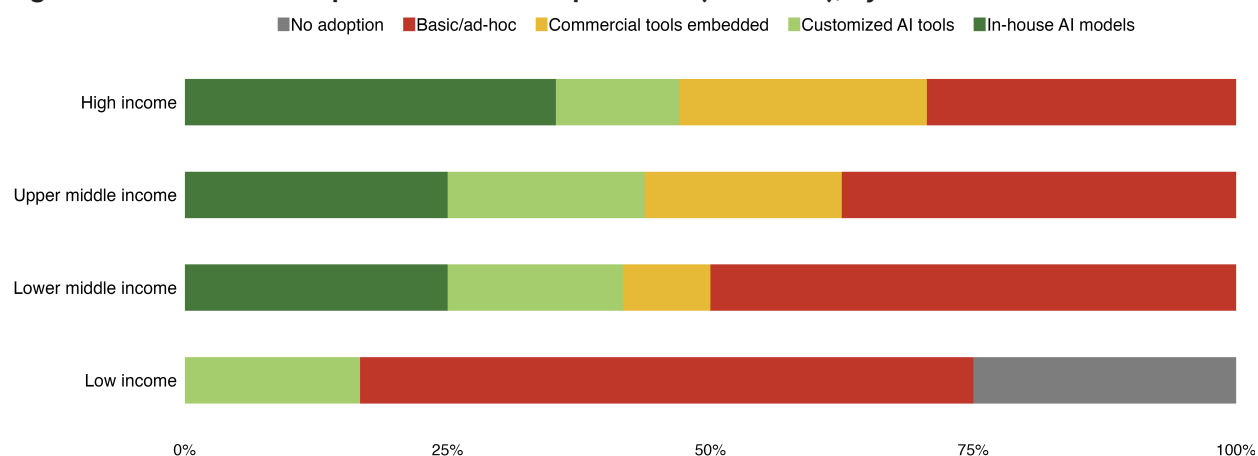
What Does AI Adoption Look Like Globally?

To put your government's responses to the AI and Data for Better Governance Survey in perspective, this section draws on data from the agency questionnaire to give a global snapshot of AI adoption, institutional readiness, and barriers.

Across all participating economies, government AI adoption is advancing but uneven. About 39 percent of surveyed governments are using customized or in-house AI tools for internal operations, while 42 percent use AI only for ad hoc tasks, such as search and summarization. Citizen-facing adoption is somewhat less advanced: only 38 percent of governments have reached an advanced stage, while 14 percent report that they do not use AI for public service delivery at all.

Levels of AI adoption vary by income level. Approximately half of high-income economies use customized or in-house AI tools for internal operations, compared with under 20 percent of low-income economies (figure 4). This variation may reflect different starting points: governments in many low-income economies are still building the data infrastructure and governance conditions that advanced AI adoption requires.

Figure 4. Levels of AI Adoption for Internal Operations (Back-End), by Income Level

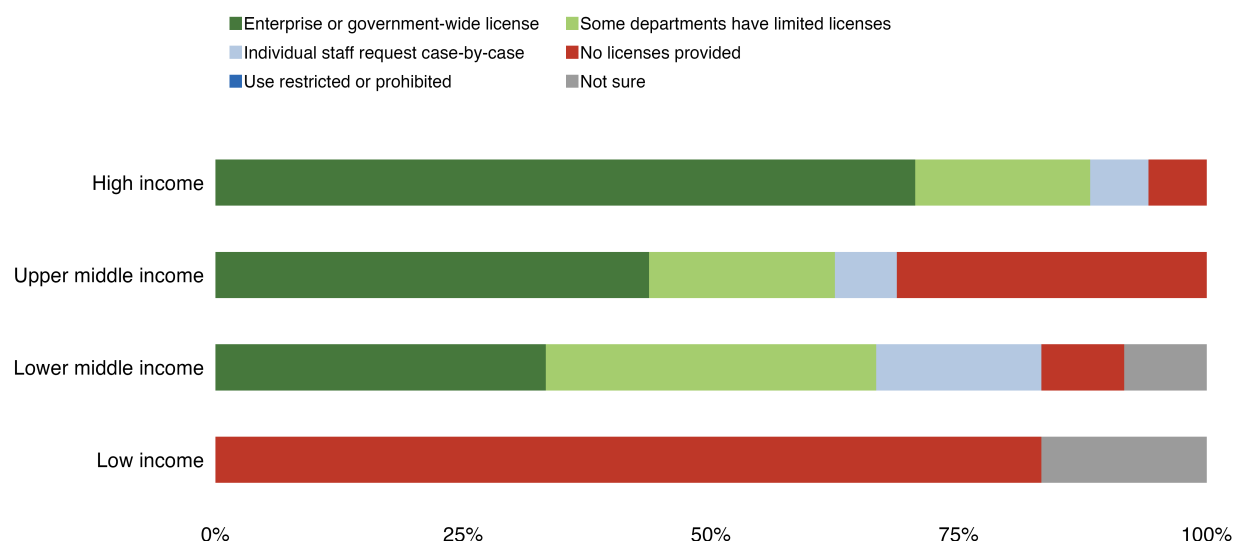


Source: AI and DfBG Survey, agency questionnaire, question 7.

AI adoption in government is moving faster than governance. Almost two in three governments provide civil servants with some form of access to GenAI tools. High-income economies are more likely to provide an enterprise or government-wide license, while low-income economies do not report providing any licenses for GenAI tools (figure 5).

On the other hand, only 39 percent of participating governments have formal, ministry-wide guidelines for AI use, 37 percent are still drafting them, and 21 percent have no guidelines. This suggests that AI access is being extended faster than the governance frameworks needed to use AI tools responsibly. High-income economies lead on both dimensions (71 percent have an enterprise license, and 76 percent have formal guidelines), and several stand out for pairing broad AI access with comprehensive governance, which covers data protection, acceptable use, prohibited use, human oversight, and vendor restrictions.

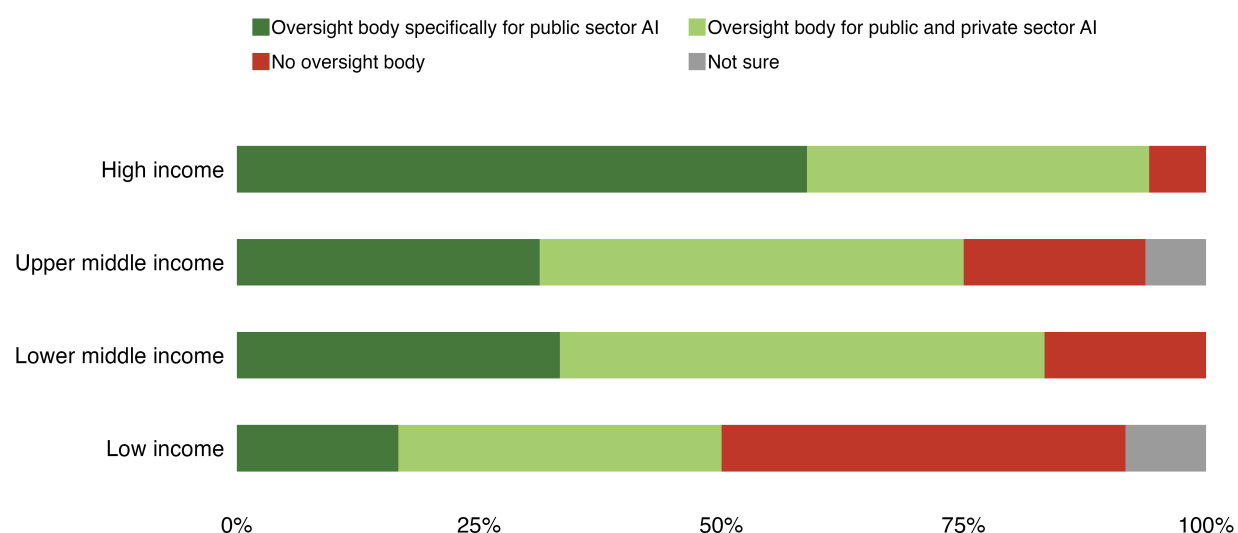
Figure 5. Licenses for Generative AI Tools, by Income Level



Source: AI and DfBG Survey, agency questionnaire, question 12.

AI oversight bodies are the most widely established governance mechanism reported in the survey: 77 percent of governments have a body responsible for overseeing AI deployment. However, bodies whose exclusive responsibility is overseeing *public* sector AI deployment and governance are more common in high-income economies (figure 6). National AI strategies are less consistently operationalized: about 60 percent of governments have adopted a strategy, but only 33 percent report that it is widely used across government. The gap between adoption and use is particularly prevalent in low- and lower-middle-income economies.

Figure 6. Type of Body Overseeing AI Deployment and Governance in the Public Sector



Source: AI and DfBG Survey, agency questionnaire, question 18.

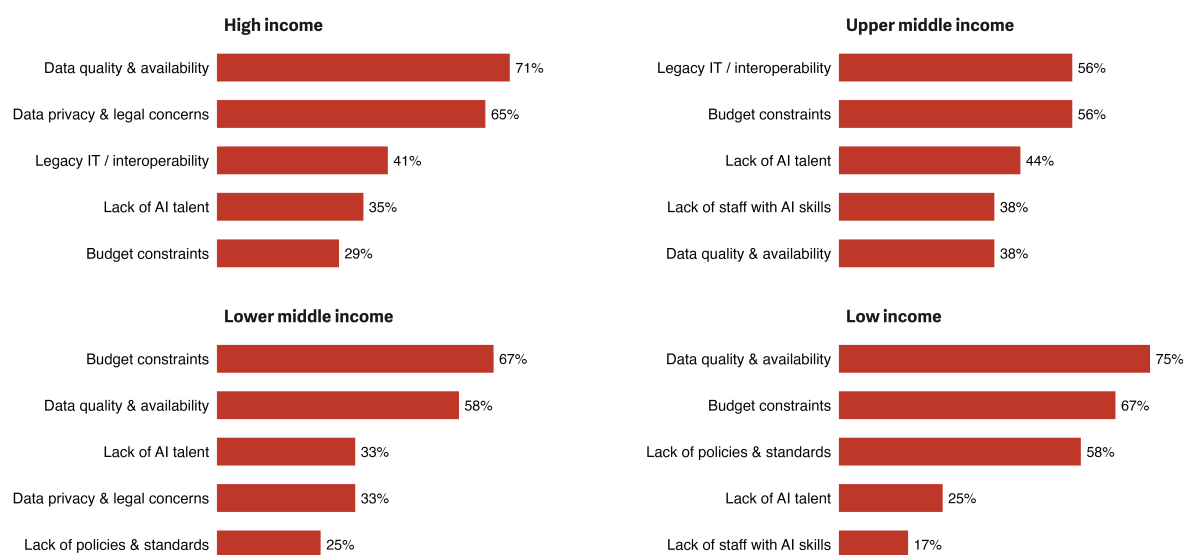
Across all income levels, governments report that data quality and availability is a key barrier to AI adoption. Roughly equal shares of low- and high-income economies report that data issues are a

barrier. Budget constraints are another common barrier for low-, lower-middle, and upper-middle-income economies. However, 54 percent of respondents report that their government has a dedicated budget for data analytics projects, and 58 percent report a dedicated AI budget. This suggests that the headline budget figures may overstate the level of sustained, multiyear investment in practice.

Other barriers vary by income level (figure 7). In lower-income economies, the lack of policies, guidelines, and standards rises as a leading barrier, reported by 58 percent of low-income and 25 percent of lower-middle-income agencies, while it does not appear among the top barriers in higher-income economies. High-income economies instead more often cite data privacy and legal concerns (65 percent). Legacy IT and interoperability also weigh more heavily in high-income and upper-middle-income economies than at the lower end. Together these patterns suggest that the barriers shift from regulatory and foundational gaps at lower income levels toward privacy, integration, and scaling concerns as capacity grows, even as data quality and budget remain common to all.

Despite these barriers, 65 percent of governments report a positive return on AI investments, and only one reports negative ones. However, several governments report that AI pilots are discontinued before they can be scaled, often because of an inadequate budget, a lack of institutional ownership beyond the pilot team, an absence of clear success metrics, or insufficient change management capacity. Apart from the technical challenges of building AI systems in the first place, this finding suggests that many governments are running successful pilots but lack the operational and institutional infrastructure to sustain them.

Figure 7. Main Barriers to AI Adoption, by Income Level



Source: AI and DfBG Survey, agency questionnaire, question 28.

What Is the Future of AI in Government?

Government is complex and highly varied, so the clearest view of AI's future comes from a nuanced approach rather than a sweeping one. The AI and Data for Better Governance Survey took that approach, focusing on specific ministries and agencies to capture both the AI applications they have already built and the ones they see as possible. Drawing on the manager and system questionnaires, the sections below trace how sectors around the world are putting government data to work today, and where they believe it could go next. We hope these examples offer a new perspective on what each sector sees in its own data and a foundation for collaboration on shared problems. We are happy to put you in touch with others in the community to learn more about specific initiatives.

How Are Different Sectors Leveraging Their MIS Data?

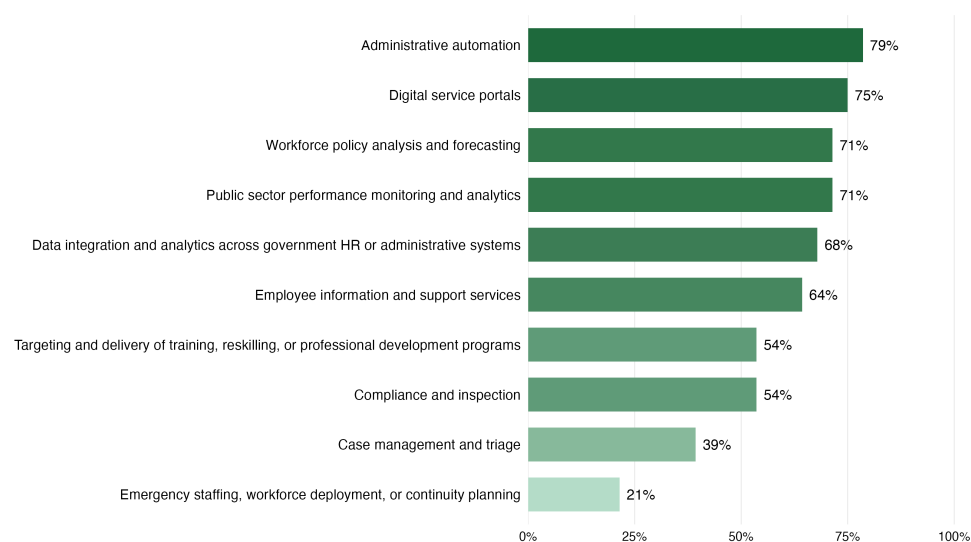
Governments across the world are finding ways to apply AI to sector-specific MIS data. What can these data be used for? The tables in the appendix collect examples from other governments' experiences. As before, these applications may not be feasible or appropriate for every context, but they give a picture of what different sectors see in their MIS data.

What Potential AI Applications Do Different Sectors See?

Beyond the AI applications they have already implemented, different sectors of government see great potential for AI to improve their performance.

The civil service sees the greatest AI potential in the operational and analytical backbone of government. Over 70 percent of respondents see potential in administrative automation (of personnel records, HR processes, payroll, and document workflows), digital service portals for recruitment, onboarding, certification, and employee self-service, and workforce policy analysis and forecasting (covering staffing needs, retirement trends, and wage bill planning). Other priorities include public sector performance monitoring and analytics and data integration across government HR and administrative systems (figure 8). Lower-priority areas include case management for grievances and disciplinary processes, and emergency staffing and workforce continuity planning.

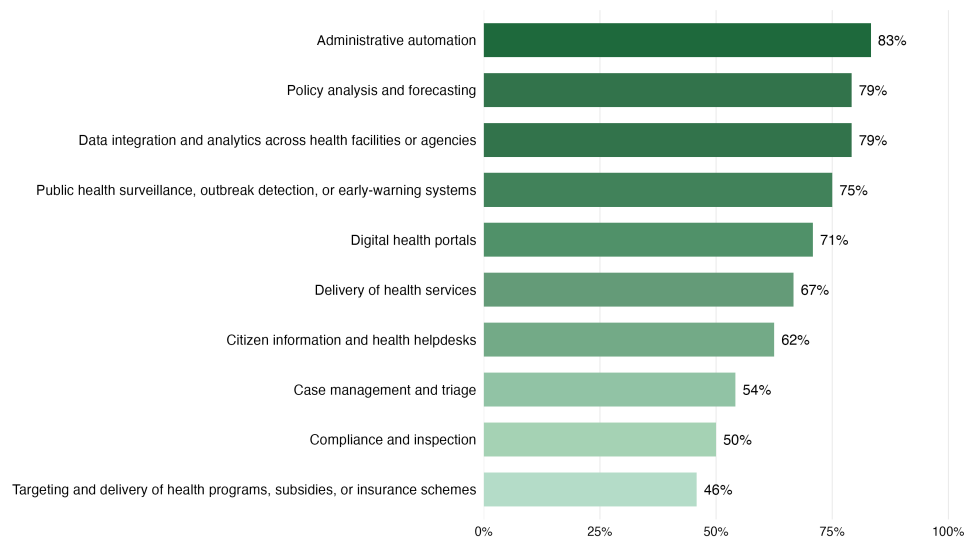
Figure 8. AI Potential in the Civil Service



Source: AI and DfBG Survey, manager questionnaire, question 22.

Health sees administrative automation (of patient records, hospital administration, and HR) and policy analysis and forecasting (focused on epidemiological trends, service demand, and budgeting) as top priorities. Respondents also emphasized public health surveillance, outbreak detection, and early-warning systems, as well as data integration across health facilities and delivery of health services through telemedicine, clinical decision support, and remote monitoring. Digital health portals for appointments, test results, electronic health records, and claims were also a priority (figure 9).

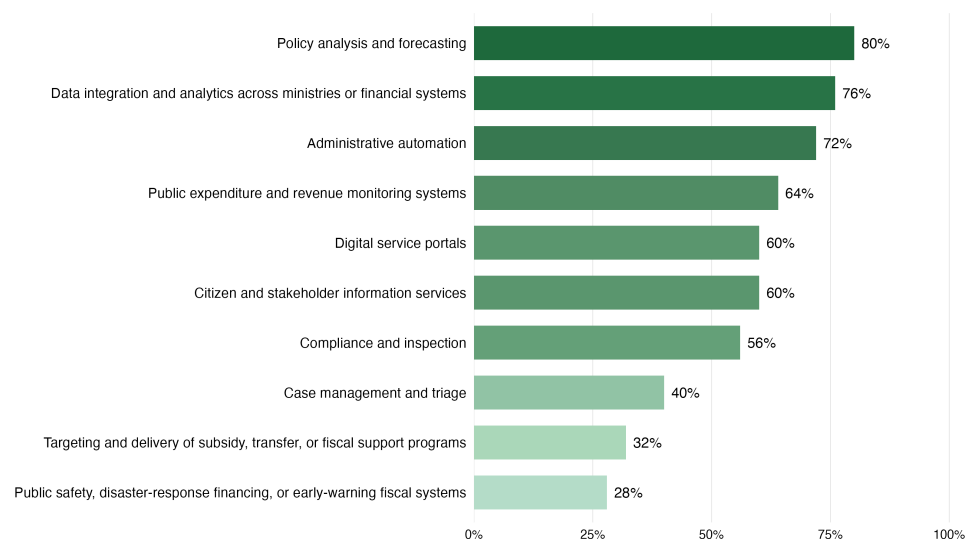
Figure 9. AI Potential in Health



Source: AI and DfBG Survey, manager questionnaire, question 22.

Public finance places policy analysis and forecasting (covering macroeconomic trends, fiscal projections, and debt sustainability) alongside data integration across ministries and financial systems and administrative automation as its top priorities. Other high-interest areas include public expenditure and revenue monitoring systems, compliance and inspection of public financial management rules, auditing and expenditure controls, digital service portals for budget submissions and financial reporting, and citizen and stakeholder information services (figure 10). Lower priorities include targeting and delivery of subsidy and transfer programs, and public safety and disaster-response financing.

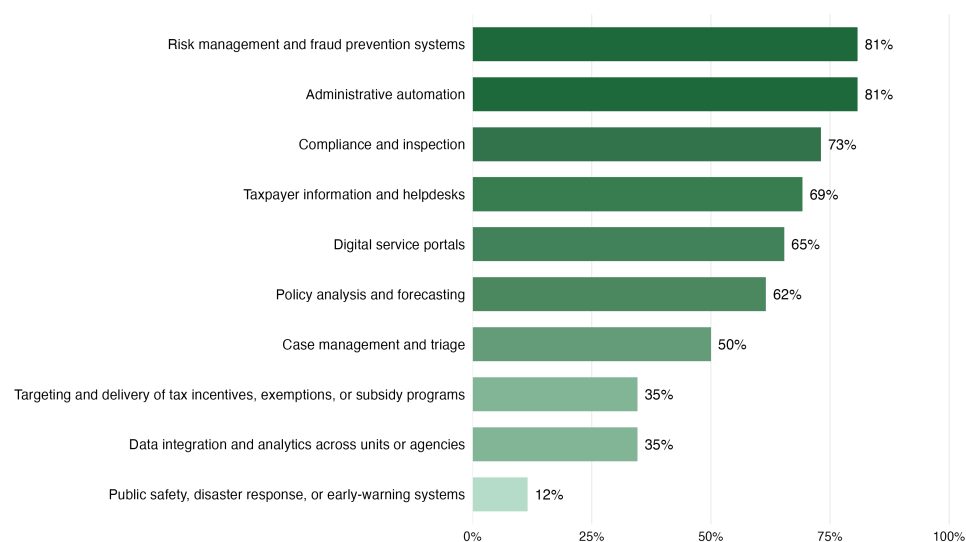
Figure 10. AI Potential in Public Finance



Source: AI and DfBG Survey, manager questionnaire, question 22.

For tax agencies, administrative automation (of taxpayer records and document processing) and compliance and inspection (covering audits, fraud detection, customs, and enforcement) are near-universal priorities. In addition, 82 percent of respondents prioritized risk management and fraud prevention systems, and 77 percent prioritized policy analysis and forecasting (of revenue projections, tax policy, and compliance trends). Other high-interest priorities include digital service portals (for tax filing, payments, refunds, and certifications) and taxpayer information helpdesks (including chatbots and hotlines) (figure 11).

Figure 11. AI Potential in Taxation

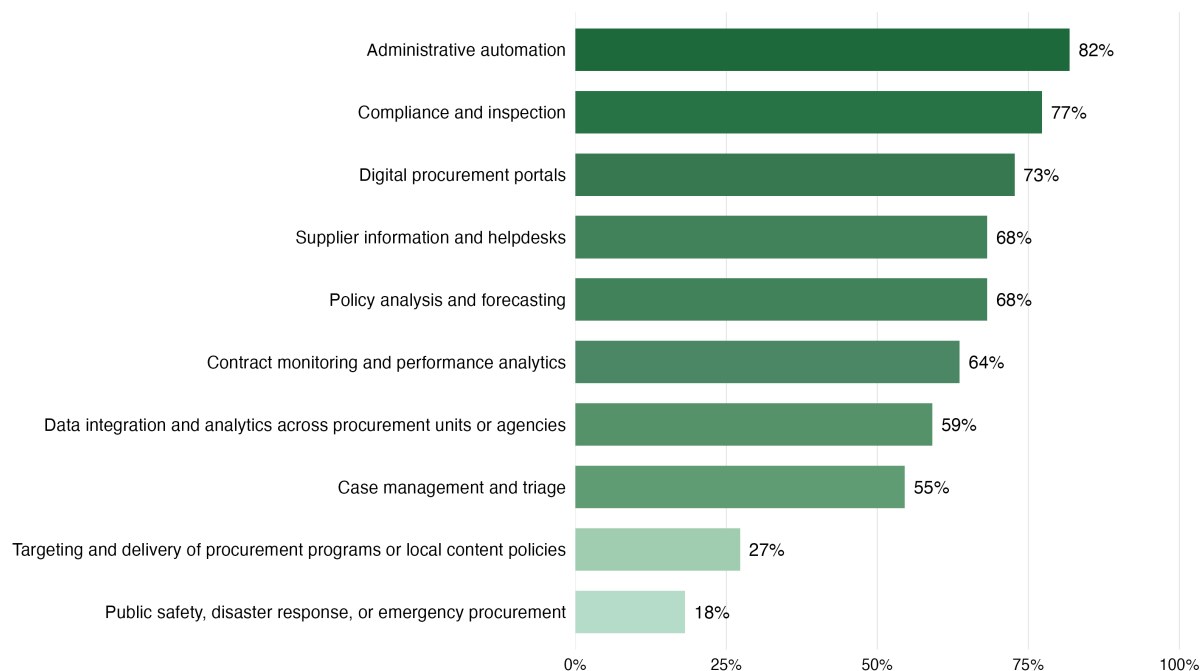


Source: AI and DfBG Survey, manager questionnaire, question 22.

Administrative automation leads as the top priority for public procurement, followed by compliance and inspection, reflecting the sector's emphasis on bid reviews, audits, and anti-corruption controls. Governments also cited policy analysis and forecasting, supplier information and helpdesks, and digital

procurement portals as the next most commonly identified potential areas, pointing to strong interest in both strategic planning and the digitization of tendering and supplier registration processes. Case management and triage, and contract monitoring and performance analytics cluster closely together in the middle range, suggesting that respondents see AI potential across the full procurement cycle rather than concentrated in any single phase. Targeting procurement programs or local content policies and public safety and emergency procurement are less commonly identified as priorities (figure 12).

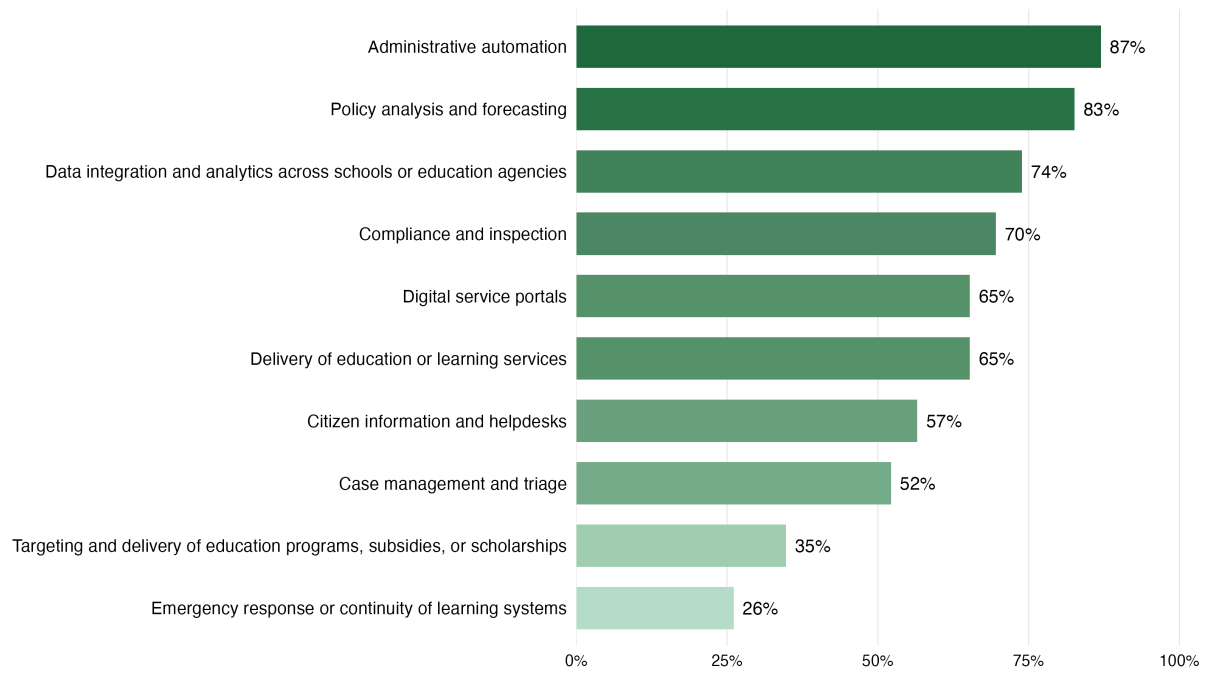
Figure 12. AI Potential in Public Procurement



Source: AI and DfBG Survey, manager questionnaire, question 22.

For the education sector, administrative automation and policy analysis and forecasting emerge as near-universal priorities, identified by most respondents. Data integration and analytics across schools or education agencies and compliance and inspection (including school accreditation, curriculum standards, and attendance monitoring) are the next most identified areas. Delivery of education and learning services (including digital classrooms, assessments, and tutoring platforms) ranks higher in education than any equivalent direct service delivery category in other sectors, reflecting the sector's distinctive relationship with end users. Digital service portals, citizen information and helpdesks, and case management for special education needs and scholarships are also priorities. Targeting of education programs and emergency response or continuity of learning systems are identified less frequently, though the latter remains a notable concern in a few economy contexts (figure 13).

Figure 13. AI Potential in Education



Source: AI and DfBG Survey, manager questionnaire, question 22.

Postscript: What Is Next for AI and Data for Better Governance?

We are grateful for your involvement in this data collection effort to understand the use of AI tools by governments worldwide, and we hope that the data reported here are useful as you map a course in the rapidly changing AI landscape. Please watch for the publication of the *World Development Report 2026: Decoding AI* later this year, which will include results and analysis of the global survey.

Surveys can only capture one moment in time, but the global landscape of AI in government is changing quickly. For this reason, the World Bank Group aims to continue collecting data on AI use in government. We welcome suggestions for how to improve the survey process.

We realize that, given the fast pace of change in this area, these results will be most useful if you are able to apply them right away to understand and respond to key problems. We are eager to use this survey as a foundation for collaboration and exchange of ideas and solutions among economies. Please reach out if we can help connect you to other members of this community to share ideas and learn from one another.

Appendix

How Are Different Sectors Leveraging Their MIS Data?

The following examples illustrate AI applications and tools that leverage MIS data across the six sectors covered by the AI and Data for Better Governance Survey from economies around the world.

TAX DATA		
Australia AI-assisted Law Interpretation and Insight AI learns from law frameworks and precedential rulings to suggest correct application of the law, helping clients comply and case officers decide consistently.	Australia MyID Fraud Detection AI scans all myID accounts to identify and risk-score the likelihood of digital identity theft (third-party fraud).	Australia Tax Time Nudging AI nudges to reduce taxpayer compliance costs and improve the efficiency of the tax system.
Austria Predictive analytics for fraud detection Predictive analytics applied to fraud detection across tax administration, including employee tax assessment and child benefit.	Cambodia BEAURAMEI (GDT Chatbot) Official digital service providing tax information, announcements, and general assistance to the public 24/7.	Canada Automated risk assessment tool Automated risk assessment in tax compliance.
Canada Genni Secure Protected B generative AI platform for personal productivity and integration into business flows, with RAG and system prompt capabilities.	Chile Tax jurisprudence chatbot Chatbot supporting access to tax jurisprudence.	Czechia POC ADAM Functional prototype of an internal chatbot with AI elements, built on selected implementation structure documents, evaluated with a focus on error rate.
Hungary CustomGPT for eVAT and Online Invoice systems Provides technical support to programmers working with the eVAT and Online Invoice systems.	Hungary Invoice item categorisation Categorises invoice items.	Hungary On-site inspection selection support Supports selection of taxpayers for on-site inspection.
Hungary Personal Income Tax Chatbot Answers questions taxpayers put to the customer service function.	Hungary X-ray image analysis for customs inspection Analysis of X-ray images to support customs inspection.	Jordan AI in declarations Artificial intelligence applied to tax declarations.
Jordan Smart assistant, Income and Sales Tax Department AI chat assistant for the Income and Sales Tax Department.	Lesotho CRM Gen AI chatbot Generative AI chatbot for customer engagement in the tax authority's CRM.	Lesotho Invoice Anomalies Detection Predictive ML model detecting fraudulent VAT invoices.
Lesotho Non-Fileers Predictor ML model predicting which taxpayers are likely not to file in the next period.	Nepal Public taxation chatbot Chatbot informing the general public about the taxation process.	Panama Anomaly detector Detects anomalies in tax data.
Panama Taxpayer risk scoring Scores taxpayers by risk level.	Peru Ordinary control selection agents AI agents reviewing ML models used for ordinary control selection,	Peru RAG for tax inspection documents

	identifying improvement opportunities and proposing algorithm corrections for data scientists to strengthen the risk management model.	Retrieval-augmented generation over institutional documents for tax inspection processes.
Peru Taxpayer appeals agent AI agent handling taxpayer challenge and appeal resources.	Philippines ORUS registration AI Online Registration and Update System using optical character recognition to process registration documents.	Philippines Revie Uses NLP to filter and categorise taxpayer queries against IRIS data, verifying user status before escalating to a human revenue officer.
Philippines Risk-Based Audit Selection System Per RMO 1-2026, identifies statistical outliers by comparing taxpayer data against industry benchmarks.	Portugal CATIA Virtual chatbot for taxpayer support.	Portugal Ebalção Inteligente (intelligent service desk) Supports triage and response at the digital service desk.
Saudi Arabia AI Product Classification System Automates HS code classification for customs.	Saudi Arabia Ask Ziyad AI digital assistant supporting ZATCA beneficiaries with zakat, tax and customs information and service guidance.	Saudi Arabia Rased Forecasts incoming and outgoing vehicle traffic at border crossings using historical, weather, seasonal and operational data, supporting port readiness, resource allocation, shift scheduling and congestion reduction.
Singapore iNAT (Intelligent Network Analysis Tool) In-house network analysis tool mapping and analyzing taxpayer relationships to strengthen compliance and investigative work.	Singapore SWEE (Smart Writing Engine for Emails) In-house generative AI writing assistant.	Singapore VICA (Virtual Intelligent Chat Assistant) AI-powered IRAS bot automating and enhancing taxpayer and public finance enquiry handling.
Thailand Nong Aree Provides tax-related inquiry services for the public.	Thailand RD-DocChat Answers questions based on relevant documents.	Thailand RD-DocGen Generates official government correspondence.
Türkiye Ozelge Identifies similar records in the ruling database, automates manual query processes, and integrates with the Revenue Administration's Ruling Commission to speed accurate responses to taxpayer requests.	Türkiye SARP Fake Document Risk Analysis Program calculating a monthly risk score for all active VAT taxpayers, profiling those who issue fake invoices using a quantitative measurement method.	Uganda Image analysis for NII Image analysis applied to non-intrusive inspection.

PUBLIC FINANCE DATA

Australia Document Register Agent (DRA) AI-enabled tool extracting and recording key information from documents in a consistent format, improving access to historical records and addressing broken links, inconsistent formats, and limited searchability.	Australia GovAI Chat Secure, government-controlled generative AI tool built for the Australian Public Service, with security, compliance, and public sector knowledge embedded in the environment.	Canada Fraud detection proof of concept Custom tool built for proof-of-concept fraud detection.
Hungary IBM Watson Analysis and prediction.	Hungary Internal ML forecasting model	Hungary

	Internally developed machine learning model for forecasting.	MiniMax accounting rule base optimisation Optimises accounting rule bases for reporting consolidation.
Peru AI platform for Conectamef AI platform supporting Conectamef.	Philippines MINAI (Managed Intelligent Network for Advanced Insight) Internal chatbot for the Department of Budget and Management, ingesting budget documents, legal documents, office orders, and circulars.	Philippines Single Content Management System with chatbot integration Proposed initiative to consolidate DBM-managed websites under a central content platform with integrated AI chatbots to improve accessibility and transparency of budget information.
Portugal Document Intelligence Assistant AI-powered assistant letting users query, analyze, and automatically summarize collections of documents with traceable sources.	Portugal S3CP Public sector expert system validating compliance of control processes to support sound public financial management, transparency, and accountability.	Singapore AGEM Audit Insights & Reporting GenAI bots detecting discrepancies in transactions, checking accuracy and completeness of documentation, and flagging non-compliance against SOPs and policy.
Singapore AI for FI Custom-built Python GenAI application letting finance officers obtain insights from finance data as natural language narrative and charts, without SQL or data analytics knowledge.	Singapore Cumulus (PRISM) ML model analysing historical patterns in supplier invoices to predict late supplier payments, enabling pre-emptive action on prompt payment KPIs.	Türkiye AI-supported accounting and advanced analytics platform Detects anomalies in public expenditure.
Uganda AI-assisted call center Generates policy-compliant responses to citizen enquiries by processing natural-language prompts against Treasury knowledge bases and workflows, routing complex questions to specialists and giving agents real-time suggested replies.	Uganda EDRMS AI document management AI-assisted document review, upload, and classification across systems, personalised search, preference-based notifications, and AI-powered retrieval and summarisation of archived policies and treasury instructions linked to source documents.	Uganda IFMS Oracle configuration support NLP agent extracting current system configuration state covering workflow, business logic, security rules, role segregation, and cross-MDA data access; analyses it to recommend workflow optimisations and security hardening; generates AI-assisted test scripts to validate configurations before production migration.

PUBLIC PROCUREMENT DATA

Benin Procurement procedures chatbot Chatbot for enquiring about procurement procedures.	Benin Submitted document analysis Analysis of documents submitted during procurement.	Benin Voice assistant Voice assistant for information intake.
Chile LLM Observatorio Detects irregularities in public procurement processes using a large language model.	Chile Price comparison Matches products across purchasing channels and the market to identify inefficiencies in public procurement.	Chile Tender evaluation agent AI agents automatically evaluate public tender offers to determine the winning bid against the selection criteria.
Czechia Zakazky.gov AI assistant for searching currently available public contracts.	Kazakhstan AI assistant for technical specifications Generates technical specifications for procurement.	Panama Bid document template generator Generates bid document templates enforcing the required technical structure within the legal context so users can focus on substantive technical detail. Pilot pending execution.

Panama Guidance agent Model trained to answer questions on public procurement processes within current regulation, intended to reduce call volume to the customer service directorate. In pilot.	Panama PREDIPEC Pre-analysis of technical criteria in the exceptional procedure review, automatically identifying presumed division of subject matter, analysing content, and identifying patterns in procurement documents to reduce manual legal analysis. Pilot pending execution.	Saudi Arabia Etimad GPT Generative AI smart assistant on the Etimad platform giving beneficiaries instant responses to inquiries and direct access to information on national public finance and procurement systems, primarily the Government Tenders and Procurement Law and its implementing regulations, from an approved knowledge base.
Singapore Category Code Recommender Automatically recommends the most relevant procurement category code.	Singapore Instruction Manual (IM) Query Chatbot Allows users to query the procurement instruction manual.	Singapore Procurement Documents Drafter Generates procurement documents such as specifications and approval of requirements from chosen source documents.
Singapore Procurement Workload Forecast Dashboard Six-month forecast of procurement workload from two to three years of past trends, helping agencies identify spikes and plan manpower.	Singapore Small Value Purchase (SVP) Assistant Helps users navigate small value procurement rules, linking to relevant guidelines, internal workflow procedures, and user guides.	Türkiye Complaint appeal summarisation Summarizes complaint appeal documents in procurement.

CIVIL SERVICE DATA

Bhutan Chatbot for RCSC Chatbot for the Royal Civil Service Commission.	Canada AIVA (AI Virtual Assistant) Automates pay intake, calculations, and document handling with outputs reviewed and approved by compensation professionals. The chatbot draws on over 80,000 business rules, 150+ collective agreements, and 40,000+ pages of directives, terms and conditions, and salary tables, shifting HR-to-pay intake from a manual model to a data-driven one.	Chile CV evaluation system Streamlines the CV evaluation stage of public service selection processes.
Chile Public records search In-depth search of public records on candidates for senior public management positions.	Chile Talent search and attraction system Talent search over a base of successful state interns, matching them to public service roles as a first job.	Jordan AI-driven question generation Generates exam and assessment questions.
Jordan Chatbot (RAG) Retrieval-augmented generation chatbot.	Kazakhstan AI model for assessing citizens' needs Model assessing the needs of citizens.	Kazakhstan AI on the Social Services Portal AI deployed within the Social Services Portal.
Kazakhstan GOV HR AI Intelligent system analyzing HR data and supporting decision-making in public sector human resource management.	Portugal Vitor Answers queries and provides information drawn from the DGAEP's FAQs.	Saudi Arabia AI Cybersecurity Advisor Automatically converts conditional rules written in different query languages into the unified query language used in the ministry's cyber threat detection systems, strengthening monitoring across employee devices and platform servers.

Saudi Arabia Establishment Verification System Uses NLP to assess whether an establishment's trade name aligns with its declared economic activity, processing unstructured text, detecting ambiguity and inconsistencies, and classifying results for regulatory reporting.	Saudi Arabia Musaned AI Interview Service Behavioral assessment of domestic workers via virtual interviews using facial expression and voice pattern analysis.	Singapore AI-powered service delivery recommender Provides personalized, proactive, omni-channel recommendations on relevant government schemes to citizens.
Singapore Job Description Generation Generates job descriptions for roles to be hired in the agency.	Singapore Orion CV Screening Screens CVs of applicants to public sector jobs.	Singapore Pair Chat Secure AI chat assistant for public officers.
Singapore Query@Gov HR Chatbot Responds to employee queries on HR policies.		

HEALTH DATA

Chile Automated clinical coding with NLP Automated clinical coding using natural language processing.	Hungary Brainomix Medical imaging platform automating validated imaging biomarkers.	Hungary Colorectal carcinoma detection AI Detection of colorectal carcinoma.
Hungary OkosSzolgáltatások (Smart Services) Middleware accessible through the eMedSolution medical system providing single-point access to AI-based services covering care, prescription, laboratory, and combined healthcare data, plus a chat function for healthcare documents.	Jordan AI Clinical Imaging Reporting (Paxera) Paxera imaging AI tools supporting radiological reporting for clinical imaging.	Jordan CAD4TB Analyzes digital chest X-rays to detect pulmonary tuberculosis.
Jordan eICU (Virtual Hospital) Virtual ICU platform with predictive algorithms identifying patients at risk of deterioration.	Mexico Breast cancer image analysis Image analysis for early breast cancer detection.	Mexico Predictive systems Predictive systems in health.
Panama Medical appointments Manages and optimises medical appointment scheduling.	Panama Preventive control of chronic diseases Preventive monitoring and control of chronic disease.	Peru AI valuation of health services AI-based valuation of health service provision.
Peru ENSAP IA Virtual assistant orienting users of the National School of Public Health (Minsa).	Peru Health service audit with machine learning Machine learning applied to auditing of health service provision.	Peru QURE IA Decision support for chest X-ray images.
Peru SIS member chatbot Chatbot serving insured members of the SIS health insurance scheme.	Philippines Administrative Issuances Processing of health-related administrative issuances with AI.	Philippines DOH Data Collect Dashboards AI-enabled analytics monitoring hospital bed capacity and supporting real-time decision making.
Philippines	Portugal	Portugal

KIRA AI-powered chatbot delivering timely and reliable health information to the public.	Detect SNS 24 Users photograph skin lesions with a smartphone; an AI model compares patterns against validated clinical image databases and assigns a risk score, enabling early identification of suspicious cases.	MODE (Waste Monitoring Model) Detects waste and fraud in healthcare expenditure using business rules combined with unsupervised decision-tree learning, analyzing billing and prescription data and assigning risk scores from 0 to 1 to patients, doctors, pharmacies, and providers.
Portugal Predictive Coding Translates electronic health record data into standardized numeric or alphanumeric formats so diagnoses and procedures can be structured for hospital financing, research, surveillance, and planning.	Portugal SICO (Death Certificate Information System) Applies AI to analyse death certificates in real time, identifying abnormal patterns in mortality data and detecting risk moments such as heatwaves or disease outbreaks so authorities can issue alerts and act preventively.	Saudi Arabia Chronic disease screening Screening for chronic disease.
Saudi Arabia Disease surveillance and outbreak forecasting Early-warning models on notifiable diseases and hospital admissions.	Saudi Arabia SEHA Virtual Hospital AI-supported virtual consultations, tele-ICU, and remote monitoring.	Saudi Arabia Sehhaty Citizen health super-app with AI features including symptom guidance and triage, personalized reminders, lab result explanations, appointment prioritization, and no-show prediction.
Saudi Arabia Tele-Stroke Platform Connects peripheral hospitals to advanced stroke centers, using AI for real-time CT analysis to accelerate diagnosis and reduce unnecessary patient transfers.	Singapore Aberration Check Detects statistical aberrations in health data.	Singapore AI 360 Regulation Supports health sector regulatory processes.
Singapore Ask MOH AI assistant answering health ministry enquiries.	Singapore Claims Fraud Detection Detects fraudulent health claims.	Thailand AI Chest X-ray Detects pulmonary tuberculosis and lung cancer from chest X-rays.
Thailand AI Retina Detects diabetic retinopathy.	Thailand Medical document summarisation Summarises medical documents.	Thailand MOR-ASR Documentation Automatic speech recognition for clinical documentation.
Türkiye AI decision support for mammography AI-assisted radiological decision support for early breast cancer detection in mammography screening.	Türkiye Brain CT Stroke AI Automatic detection of stroke, hemorrhage or occlusion, from CT images.	Türkiye MKYS AI Project Extracts product attributes from free-text medical supply records, standardises the data, and detects duplicate entries to improve data quality.
Ukraine AI analysis of CT scans AI-based solutions in healthcare facilities enabling faster and more accurate assessment of CT results.	Ukraine AI diagnosis of tuberculosis and lung conditions AI supporting diagnosis of tuberculosis and other lung conditions.	Ukraine Digital vision screening Early detection of retinal damage in patients with diabetes to ensure timely ophthalmologist referral and reduce risk of vision loss.

EDUCATION DATA

Greece	Greece School vocational guidance	Guatemala Administrative procedures chatbot
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AI assistant of the Panhellenic Schools Network AI assistant serving the national schools network.	Vocational guidance for school students using machine learning techniques.	Chatbot answering queries on simplified administrative procedures in natural language.
Guatemala Natural language queries of education registries Natural language querying of student and teacher registry systems.	Jordan Siraj Platform for students and teachers.	Kazakhstan AI expert for educational programs Expert system evaluating and advising on educational programmes.
Kazakhstan Data Integrity AI AI applied to data integrity in education systems.	Kazakhstan Risk Identification Agent (RegTech AI) Identifies risks in educational regulation.	Kosovo MIA (Learning with Artificial Intelligence) Supports students preparing for the Achievement Test, Matura Exam, and international assessments; teachers in lesson planning, materials development, and assessment; and institutions in digitalisation and quality assurance. Operates in standard Albanian and adapts to user role and level.
Peru Classroom aggressive behaviour identification model Model identifying aggressive behavior in the classroom.	Peru Peru Educa Platform aimed at improving learning outcomes in Peru.	Portugal iaedu.pt Makes professional large language models available to higher education institutions.
Saudi Arabia Generative AI through Madrasati Enables teachers and students to use generative AI tools through the Madrasati educational account for content creation, assessment design, classroom support, and learning activities.	Saudi Arabia Madrasati AI Assistant Automated responses to beneficiary inquiries, supporting access to back-to-school and Madrasati platform services.	Saudi Arabia Student Admission prediction Predicts student admission outcomes and supports admission-related decision-making through predictive analytics.
Singapore AI Platform Evaluation of AI trust, safety, and accuracy.	Singapore LangBuddy Interactive mother tongue language voice chatbot.	Singapore LEA (Learning Assistant) Student tutoring.
Singapore Markly AI feedback and marking assistant.	Türkiye DİLİM AI-supported language learning platform aligned with CEFR, assessing learner performance in speaking and writing tasks and providing targeted adaptive feedback.	Türkiye EBA Assistant LLM-based personalized learning assistant delivering customized content recommendations, study plans, and instant academic support, using generative AI and learning analytics.
Türkiye MEBBYS AI-supported decision-support and ministry intelligence system integrated with GIS, using ML on large-scale educational, administrative, and spatial data for predictive analytics, early warning, and real-time insights supporting strategic planning and resource allocation.	Türkiye MEBİ Individual Learning Platform Digital education ecosystem providing personalized, AI-supported learning for students and graduates.	Türkiye Personalized guidance robot for lifelong learners Chatbot directing trainees to suitable training programs based on age, education level, city of residence, and prior learning.
Uruguay	Uruguay	Uruguay

Adaptive English Test Adaptive test adjusting to learner performance.	Aleks Adaptive mathematics platform.	OBI chatbot Student guidance on executive functions.
Uruguay PowerBuddy Content generation for teachers, embedded in the LMS.	Uruguay SAT tutors Supports marking of assignments.	Uruguay School dropout forecasting Forecasts which students are at risk of dropping out.

Source: AI and DfBG Survey, system questionnaire, question 13, and manager questionnaire, question 11.